

AI Resume Analyzer

Project Documentation

1. Problem Statement

Job seekers struggle to tailor resumes for each job description. Most resumes are rejected by Applicant Tracking Systems (ATS) due to missing keywords or poor alignment with job requirements. Existing tools are either expensive, manual, or lack structured feedback, making the process time-consuming and inefficient.

2. Solution Overview

AI Resume Analyzer is a web application that compares a user's resume with a job description and provides a match score, keyword analysis, gaps, and actionable recommendations. The goal is to help users quickly optimize resumes for better chances of passing ATS screening.

3. Tech Stack

Frontend: • React – UI development • TypeScript – type safety • Vite – build tool • Tailwind CSS – styling AI & APIs: • Groq API (Llama 3.3 70B) – main analysis engine • Google Gemini – fallback AI model File Processing: • pdfjs-dist – PDF parsing • mammoth – DOCX parsing Deployment: • Vercel / Netlify – hosting • GitHub – version control

4. Key Features

- Resume and job description upload (PDF, DOCX, TXT)
- AI-based match score (0–100)
- Keyword analysis (matched vs missing)
- Gap detection between resume and job description
- Actionable improvement suggestions

5. Outcome

The project demonstrates how AI can automate resume evaluation and improve job application efficiency. It provides fast, structured, and user-friendly feedback, reducing the time required for manual resume customization.